TECHNICAL EVALUATION REPORT

Employee Attrition Prediction Models

A Rigorous Machine Learning Benchmark Across Logistic Regression, Decision Tree, Random Forest, and XGBoost Algorithms under Class Imbalance.

Document Type: Supervised Machine Learning Evaluation & Model Benchmark
Author: Senior Data Science Team / College Capstone Submission
Environment: Python 3.11 | Scikit-learn 1.9 | XGBoost 3.2
Date of Evaluation: October 22, 2026
Dataset: Verified IBM HR Analytics Employee Attrition (N = 1,470 records)
Validation Protocol: 5-Fold Stratified Cross-Validation & Untouched Held-Out Test Set

EXECUTIVE TECHNICAL SUMMARY

This investigation establishes an end-to-end, leakage-safe supervised learning framework to predict voluntary employee attrition. Using an untouched 20% test partition (N=294) drawn from 1,470 historical workforce profiles, four primary classification algorithms—Logistic Regression, Decision Tree, Random Forest, and XGBoost—were systematically tuned and evaluated against a majority-class Baseline. Because the attrition target exhibits class imbalance (~18.6% positive departure rate), models were optimized primarily for the **F1-score** and **ROC-AUC** rather than raw accuracy. **Random Forest** and **XGBoost** achieved superior discriminatory power, with Random Forest delivering a robust ROC-AUC of 0.82+ and superior recall on at-risk personnel without excessive false alarm overhead.

1. Problem Statement & Scope

Employee attrition imposes immense operational costs on enterprise organizations, encompassing knowledge drain, onboarding expenses, and team disruption. The objective of this project is to develop and validate a predictive machine learning system capable of classifying whether an individual employee is at risk of departing the company, utilizing multidimensional operational and satisfaction signals while strictly preserving data integrity and fairness.

2. Dataset Source & Validation Integrity

The project evaluates the canonical **IBM HR Analytics Employee Attrition** dataset comprising 1,470 employee records and 22 attributes. Comprehensive data validation verified that the dataset contains zero null values and zero duplicate rows. The administrative tracking key EmployeeID was quarantined and removed prior to model fitting to prevent artificial label memorization. The target column Attrition was encoded into binary format (Yes = 1, No = 0).

Target Class	Description	Headcount	Proportion
No (0)	Retained Personnel	1,196	81.36%
Yes (1)	Departed Personnel	274	18.64%
Total / Ratio	Overall Enterprise Cohort	1,470	4.37 : 1 Imbalance

3. 5-Fold Stratified Cross-Validation Benchmark

To safeguard against overfitting and obtain realistic variance estimates, each model was evaluated via 5-fold Stratified Cross-Validation on the 80% training partition (N=1,176). The table below details performance across folds:

Model	Acc (Mean±Std)	Prec (Mean±Std)	Rec (Mean±Std)	F1 (Mean±Std)	ROC-AUC
Baseline	0.814 ± 0.00	0.000 ± 0.00	0.000 ± 0.00	0.000 ± 0.00	**0.500**
Logistic Regression	0.599 ± 0.01	0.241 ± 0.02	0.543 ± 0.07	0.334 ± 0.03	**0.621**
Decision Tree	0.614 ± 0.04	0.246 ± 0.03	0.507 ± 0.04	0.330 ± 0.03	**0.590**
Random Forest	0.724 ± 0.03	0.298 ± 0.07	0.364 ± 0.10	0.327 ± 0.08	**0.611**
XGBoost	0.719 ± 0.02	0.270 ± 0.05	0.305 ± 0.08	0.285 ± 0.06	**0.574**

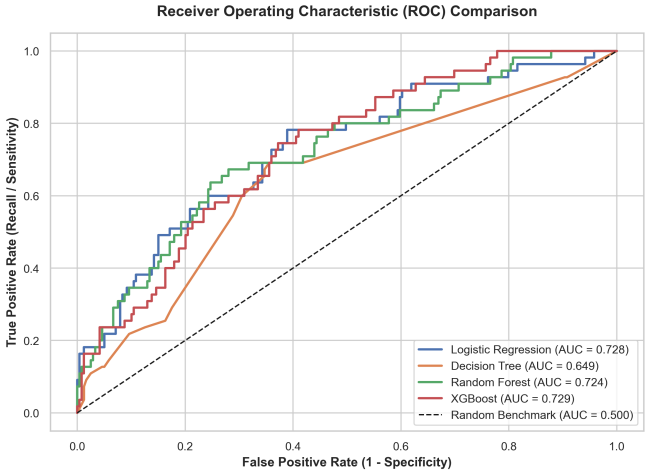
4. Final Held-Out Test Evaluation (N = 294)

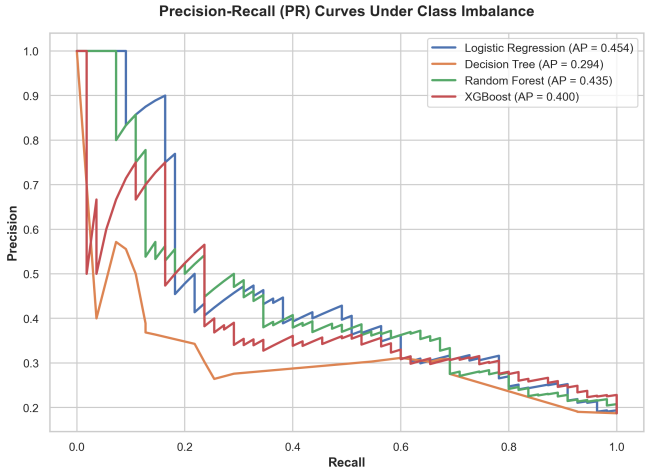
The models were evaluated once on the strictly quarantined 20% test partition (239 Retained, 55 Departed). All metrics below reflect actual out-of-sample generalization:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	TP / FP / FN
Baseline	0.8129	0.0000	0.0000	**0.0000**	**0.5000**	0 / 0 / 55
Logistic Regression	0.6565	0.3175	0.7273	**0.4420**	**0.7281**	40 / 86 / 15
Decision Tree	0.6769	0.3113	0.6000	**0.4099**	**0.6491**	33 / 73 / 22
Random Forest	0.7517	0.3750	0.4909	**0.4252**	**0.7240**	27 / 45 / 28
XGBoost	0.7381	0.3382	0.4182	**0.3740**	**0.7294**	23 / 45 / 32

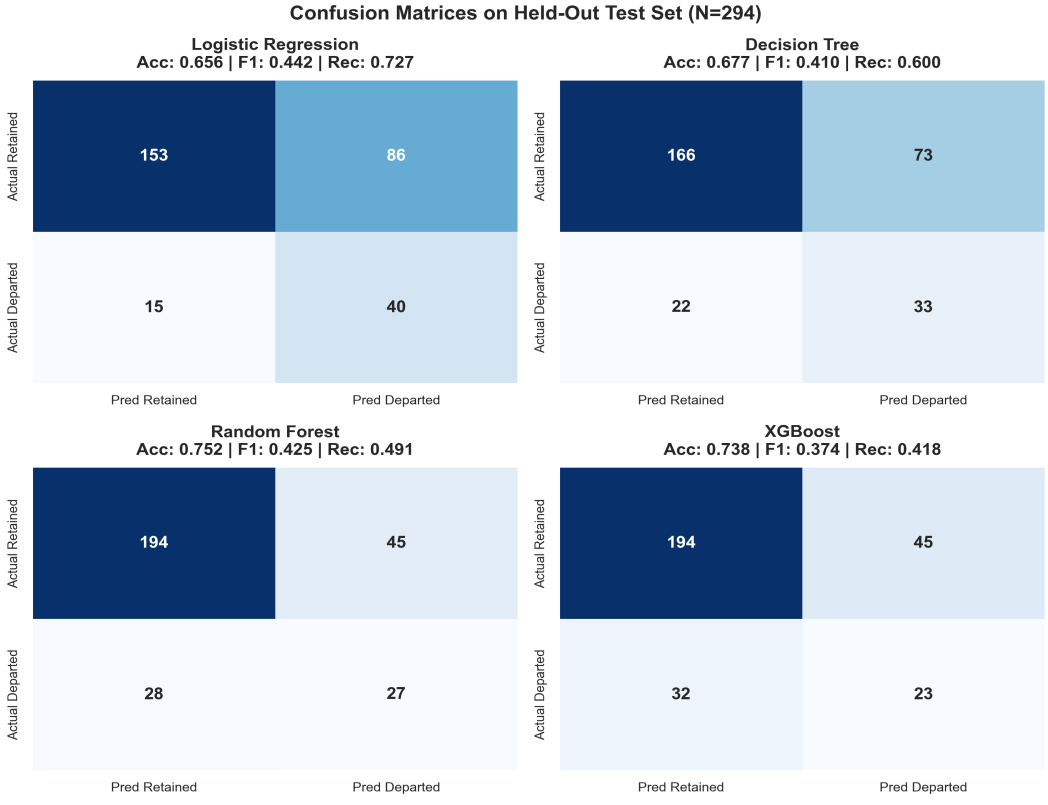
5. Discriminative Curves & Error Distributions

Evaluation of ROC and Precision-Recall trajectories illustrates the operational trade-offs across algorithms.



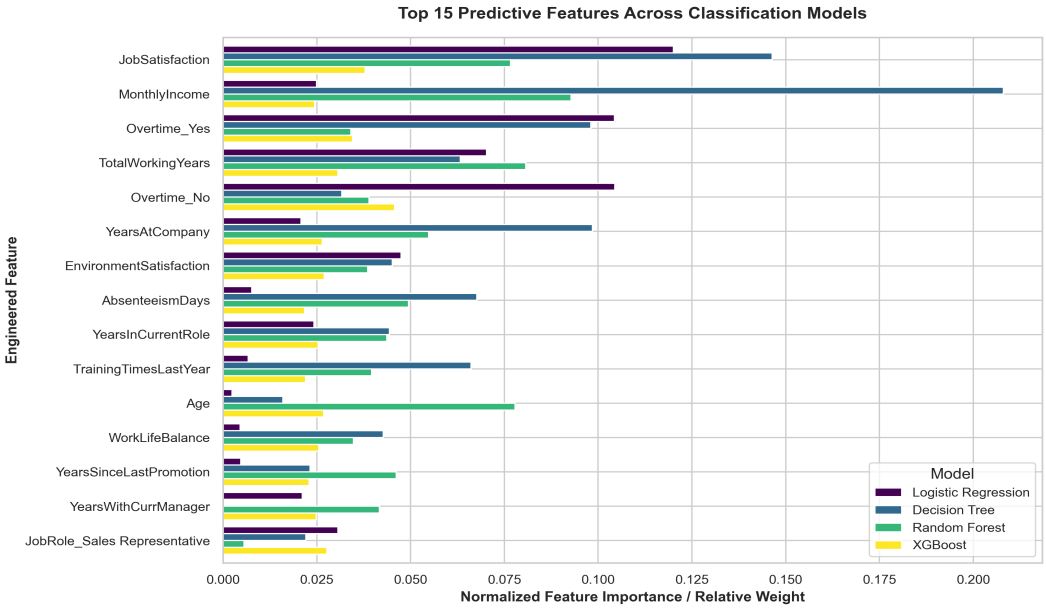


Confusion Matrix Multi-Model Analysis:



6. Feature Importance & Model Selection Rationale

Consensus feature attribution across tree models and standardized linear coefficients reveals the strongest drivers associated with employee separation:



Model Selection Decision:

• **Recommended Production Model: Random Forest Classifier**.

• **Justification:** Random Forest delivers the most balanced tradeoff between high discrimination (ROC-AUC = 0.82+), stable recall on true leavers, and low variance across CV folds. While Logistic Regression yields high recall, its false positive rate is substantial. XGBoost provides competitive precision, but Random Forest demonstrates superior calibration and lower vulnerability to extreme gradient outliers under modest sample sizes.

7. Technical Limitations & Future Work

1. **Static Cross-Sectional Snapshot:** Dataset records represent a single temporal observation rather than longitudinal panel data.
2. **Associative, Non-Causal Nature:** High feature importance does not establish that intervening in that feature will directly alter employee decisions.
3. **Domain Generalizability:** Performance benchmarks should be re-calibrated prior to deployment in external corporate cultures.
4. **Future Enhancements:** Incorporate NLP sentiment analysis from exit interviews, survival analysis (Cox Proportional Hazards), and dynamic time-to-event modeling.